

RAKESH KUMAR

Software Developer | Full-Stack & Backend Systems

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PROFESSIONAL SUMMARY

Software developer specializing in backend and full-stack systems, with an incident root-cause engine that achieved 89.5% accuracy, 16 percentage points above a single-prompt LLM baseline. Other projects include a constraint solver that found a 46-session schedule with zero backtracks and a real-time collaborative code editor with CRDT synchronization; both are deployed and demoable.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++, SQL
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT, RBAC
Frontend: React.js, Next.js, D3.js, Tailwind CSS, HTML5, CSS3
Databases & ORM: PostgreSQL, MongoDB, MySQL, Redis, pgvector, Prisma, SQLAlchemy
AI / LLM: Gemini API, Vector Embeddings, LLM Evaluation
Systems: Yjs (CRDT), WebAssembly, Web Workers
Tools, Testing & Deployment: Git, GitHub, Docker, Postman, Vitest, Pytest, Supertest, Vercel, Render

PROFESSIONAL EXPERIENCE

Software Development Intern - Codetech IT Solutions | Remote

Jan 2026 – Apr 2026 | Node.js | Express.js | MongoDB | JWT | RBAC | REST APIs | Git | Postman

- Built a role-aware employee management system that replaced spreadsheet-based tracking and supported employee, manager, and admin workflows.
- Secured CRUD REST APIs for employee, department, and role modules with JWT authentication, RBAC middleware, and request validation; used Postman to test operations and validate responses.

PROJECTS

TRACE - Telemetry Root-Cause Autonomous Critique Engine | [GitHub](#) | [Live](#)

Python | FastAPI | PostgreSQL (pgvector) | Gemini API | Next.js | TypeScript | D3.js

- Achieved 89.5% root-cause accuracy across 19 synthetic incidents with hidden root causes, versus a 73.7% single-prompt LLM baseline, using deterministic scoring and an LLM critique loop.
- Correlated logs, metrics, traces, and deployment events using temporal and causal evidence; added a deterministic trend-comparison check after the engine incorrectly attributed a memory leak to a recent deployment.

CHRONOS - Constraint-Based Timetable Scheduling Engine | [GitHub](#) | [Live](#)

React | Node.js | PostgreSQL | Gemini API | D3.js | GSAP

- Found a valid 46-session schedule in 46 search nodes with zero backtracks using MRV/LCV; chronological search found no solution within a 10-million-backtrack budget on the same input.
- Built a backtracking constraint solver with live D3.js search visualization; integrated Gemini API to translate natural-language scheduling requirements into structured constraints.

SyncPad - Real-Time Collaborative Code Editor | [GitHub](#) | [Live](#)

React | TypeScript | Node.js | Yjs (CRDT) | Monaco Editor | WebSocket | Pyodide (WASM)

- Enabled conflict-free concurrent editing with Yjs CRDT synchronization, live cursor sharing, and collaborator presence.
- Implemented browser-based execution for JavaScript, TypeScript (via transpilation), and Python using Web Workers and Pyodide (WebAssembly), keeping execution off the main UI thread.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science Engineering | Graduated: May 2026

Parul University, Vadodara, Gujarat

CERTIFICATIONS & ACTIVITIES

- SQL (Advanced) – HackerRank
- C++ Programming – OpenEDG / Cisco
- Introduction to IoT (Elite) – NPTEL (IIT Kharagpur)
- AMENTIS Hackathon – IEEE GTBIT
- CodeKshetra Coding Contest – GeeksforGeeks (GD Goenka)